

Co-Designing Emotion: Supporting Positive Affect in a Culturally Sensitive Application for African-American Heart Health

Sowmya Chandra
sowchan@iu.edu
Indiana University - Indianapolis
Indianapolis, Indiana, USA

Valerie Thompson
vthomp21@jhmi.edu
Johns Hopkins University
Baltimore, Maryland, USA

Grecia B Vargas Meléndez
gvargas4@jhmi.edu
Johns Hopkins University
Baltimore, Maryland, USA

Tanjala S. Purnell
tpurnell1@jhmi.edu
Johns Hopkins University
Baltimore, Maryland, USA

Aqueasha Martin-Hammond
aqumarti@iupui.edu
Indiana University - Indianapolis
Indianapolis, Indiana, USA

ABSTRACT

Heart disease is a leading cause of injury and death among African Americans in the United States. Culturally sensitive design of health technologies can be useful for motivating health behavior change, especially in underserved communities. Yet, designing culturally sensitive technologies requires careful consideration to identify appropriate approaches for tailoring applications to the needs of the target population. In the paper, we explore the relationship between user values and emotions for informing the design of a culturally sensitive heart health application for African Americans. We partner with members and local community advocates involved in an existing evidenced-based community heart health intervention to understand their values and explore implications for designing for positive emotional experiences in a future digital application. We contribute a discussion of emotional intentions important to the design of a future application and suggestions for designers of similar applications.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → **Health informatics**; **Health care information systems**.

KEYWORDS

culturally sensitive design, community-based design, co-design, emotion

ACM Reference Format:

Sowmya Chandra, Valerie Thompson, Grecia B Vargas Meléndez, Tanjala S. Purnell, and Aqueasha Martin-Hammond. 2024. Co-Designing Emotion: Supporting Positive Affect in a Culturally Sensitive Application for African-American Heart Health. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3613905.3650893>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650893>

1 INTRODUCTION

Heart (Cardiovascular) disease is a leading cause of injury and death among African Americans in the United States, often due to racial disparities such as limited access to care or access to effective educational resources [28]. Prior work suggests that to reduce health disparities, tailored interventions [28], such as tailored health technologies, can be one useful tool for mobilizing the African-American community by providing opportunities for scaling community programs [17, 26] and improving health outcomes [5, 6].

Culturally sensitive design can be a valuable approach for motivating health behavior change among underserved communities [3, 20, 21]. Therefore, health technologies that employ a culturally sensitive approach to design have shown the potential to positively influence user engagement with those technologies [11, 19, 22, 23, 25, 30]. Culturally sensitive design broadly emphasizes accounting for cultural sensitivities and values in design and the design process [13]. Yet designing culturally sensitive technologies requires careful consideration and tailoring [11, 14, 32] to overcome potential design tensions arising from varied perspectives and cultural differences even among similar groups.

This study aims to identify implications for designing a culturally sensitive digital health intervention for African American heart health. In particular, the study aims to identify the emotional intentions [37] important to evoke in the design of the digital application by exploring what users value in the existing program and the relationship between those values and their design preferences. For this study, we partner with members and local community advocates involved in an existing 20-year-old (non-digital) evidenced-based community program that uses stories of the Underground Railroad [1] to motivate users to participate in heart-healthy behaviors [26].

We conducted a workshop with 11 prior program members and other local community advocates to understand their values related to their experience in the program. We craft and leverage mood boards [8] as a tool for exploring the types of emotional experiences participants desire in a digital application of the intervention by identifying keywords representing emotional experiences they value. We found that participants of the existing program valued most being able to learn about their cultural and ancestral history while participating in what they described as a uniquely supportive community of people on a shared health journey. Because of their values, they preferred mood board designs emphasizing "learning and education" and "playful and game-like" experiences. Apart

from a pleasurable experience [37], participants valued culture and community, and we identified associated keywords important for evoking a positive experience in a future digital application design. This work builds on prior efforts that explore emotion in the design of user experiences [18, 34, 37], and ways to identify emotional intention [37] to guide the design process.

2 BACKGROUND AND RELATED WORK

We discuss the role of culturally sensitive design of health technologies to motivate behavior change and the role of emotion in design.

2.1 Culturally Sensitive Health Technologies to Motivate Behavior Change

Researchers have found that culturally sensitive approaches to designing health technology can be an effective way of motivating different underserved populations [3, 20, 21] and in parallel, not considering culture can lead to additional health disparities [3]. Several researchers have studied the impact of culturally sensitive design on the motivation and health behaviors of different groups and have found that tailoring health technologies can be beneficial for promoting positive affect [11, 19, 22, 23, 25, 30]. In their work, they highlight the importance of understanding the broader socio-technical context [14] for which technologies are designed, designing "deeply local" applications [11], and understanding nuanced perspectives and cultural differences even among seemingly similar groups [32]. Therefore tailoring can be important for improving behaviors, especially among underserved populations [20]. Yet designing culturally sensitive technologies requires careful consideration of the needs of those involved to identify their values [11, 14, 32] as different perspectives and cultural nuances may exist even among similar groups.

2.2 Emotion and Design

Brave and Nass note that while definitions of emotion differ, there are common aspects that contribute to the definition, including "(a) a reaction to events deemed relevant to the needs, goals, or concerns of an individual" and "(b) physiological, affective, behavioral, and cognitive components" [4]. It is well-known that emotion plays an important role in the design of user experience [18]. For example, McCarthy and Wright highlight emotion as one of four threads of user experience and argue that the emotional quality of an experience, whether satisfying or frustrating, tends to set the tone for the overall experience [18]. Therefore, within the Human-Computer Interaction (HCI) community, there has been a lot of emphasis on emotion and the role it plays in design. Studies have explored how people use technology to regulate emotion [29, 31, 35], how technologies manipulate users' emotions [7], and the emotional work tied to researching different technologies [36]. Therefore, recent work calls for more exploration of the relationship between emotion and technology [34].

This paper builds on work that aims to explore how design can be used to support positive emotional experiences by exploring the interplay between values and emotion in culturally sensitive design. Yoon and colleagues explored how to design for emotion beyond pleasure using a proposed appraisal-based emotion-driven

design approach that emphasized determining emotional intentions [37]. In the initial stages of this process, activities focus on identifying positive emotions to evoke (e.g., pride, relief, satisfaction) and exploring or "appraising" the concerns tied to those emotions. They argue that through this process, a design team can identify concerns and qualities important to interaction design [37]. We leverage this approach to understand what members of the existing program value in the current intervention to uncover emotional intentions important to designing a future culturally sensitive digital application designed to support the same goals.

3 CO-DESIGN WORKSHOP

To prepare for the co-design workshop, the design team led several rounds of iterative brainstorming to craft ideas about the storytelling and cultural aspects of the digital application design. Each round of brainstorming included research team members from the health team who were familiar with the existing program goals and the design team familiar with Human-Computer Interaction (HCI) and design processes. Mood boards are often a communication tool that is represented as a collection of images brought together to elicit sensory and emotional responses when engaging in design discussions [8]. Mood boards have a long history in design, mainly as a tool to aid designers' creative thinking but also as a communication [8] and research tools [8, 16] to explore the emotional aspects of design. Our main reason for using mood boards aligned with goals for exploring participants' desired user experience in a digital version of the intervention. However, we also wanted to intentionally explore the emotional aspects of design, given the potential of the subject matter, the Underground Railroad, to evoke potential positive (e.g., stories of freedom and resilience) and negative (e.g., past traumas) emotions among African Americans.

3.1 Mood Board Design

In brainstorming sessions, the team discussed different aspects of that story that could be used as inspiration for the design. Based on these discussions, the team crafted three initial mood boards to share with participants in the co-design workshop.



Figure 1: Examples of the types of images and colors included in the (a) Historical (b) Darkness, and (c) Gamify-Playful mood boards presented during the workshop.

3.1.1 Mood Board 1: Historical. The first mood board, *Historical*, was directly inspired by the design materials used in the existing program (Fig. 1). The program components included real-life historical documents and stories in materials to inspire participants to engage in weekly educational content. The mood board included images of historical artifacts and like-life images of abolitionists

such as Harriet Tubman that showed aging and weathering over time. We proposed shades of brown, taupe, and green and Script and Serif typography often seen in historical documents and older maps. The mood was modeled after that of a museum experience, aiming for participants to feel as if they were browsing different cultural artifacts and documents.

3.1.2 Mood Board 2: Darkness. The second mood board, *Darkness*, was inspired by games such as *Oregon Trail*, which were designed to educate others about life in another period (Fig. 1). It was designed to highlight the conditions freedom seekers might have experienced on their journey. Since the existing program used the metaphor of obtaining "*health freedom*" as one of its mottos, we brainstormed ideas related to the group experiencing the journey together as a group of freedom seekers overcoming darkness to reach freedom (the light). The mood board, therefore, included dark tones of black and gray and "scary" images to represent the wilderness at night. We also included Serif and Decorative fonts for typography.

3.1.3 Mood Board 3: Gamify - Playful. The third mood board, *Gamify - Playful*, deviated from the emphasis on history and the conditions participants likely experienced on the Underground Railroad and focused on emphasizing the programs' emphasis on togetherness and a shared journey (Fig. 1). To do so, we envisioned a game-like aesthetic with light, playful fonts and colors such as pastels and dimmed brights. Because we wanted to emphasize the focus of the journey, the mood board also included images of different gaming maps. Typography included playful Decorative and Sans Serif fonts.

3.2 Workshop Procedures

We conducted a virtual co-design workshop where we partnered with eleven prior members of the community-based health program and other local community advocates. In total 15 people attended the workshop, including four members of the research team (three with expertise in HCI and one with expertise in Public Health/Health Equity). The broad goal of the workshop was to understand existing values and obtain initial feedback from participants on how to represent those values within the future application. The design goal, therefore, aimed to identify values, understand needs and desires for motivational support, and uncover aspects of design that promote positive emotion when using the application. We aimed to answer the following research questions: (1) *What did participants value in the existing community-based program?* (2) *What is the relationship between existing user values and their preferences for design?* (3) *What emotional intentions are important to evoke in the future application design?*

During the workshop, two researchers facilitated conversations, and another two managed the collaborative technology, took notes, and recorded them on a shared discussion board. We began the workshop by setting the tone for the co-design session as a place where all voices are valued. We began the session by asking participants and researchers to introduce themselves and share their roles in the project or prior program. We followed by presenting some findings from our preliminary work with the same group. We presented preliminary findings to ensure that our interpretations

of their needs were valid and invite discussion of things we may have missed.

For the remainder of the workshop, we invited open discussion, asking participants to reflect and share their experiences in the program, what they valued in the current program, and things that they would like to see improved. We presented each initial mood board to participants, discussed the inspiration and goals, and provided an overview of the images similar to as discussed in Section 3.1. We asked the participants to reflect and provide a discussion of the different mood board elements and elaborate on whether they felt the aesthetic would be appropriate for motivating them to engage in the different components of the intervention (e.g., tracking exercise, food, or health indicators). We also asked them to comment on how the different elements of the mood boards might align with their values. Therefore, the primary use of the mood board was as a probe [15] to support discussions around participants' values in design. The session lasted approximately 2 hours and was video recorded with participants' verbal consent.

Following the workshop, the video was manually transcribed by one of the research team members. Two researchers independently analyzed the transcripts using constant comparative methods [10] to draw out codes and themes.

4 FINDINGS

To provide context and highlight concerns, we first share findings on what participants valued in the existing community-based program. We share feedback they provided on the mood boards, their preferences, and how they aligned with the values they held related to the existing program. We end by discussing the emotional intentions as defined by [37] that we identified representing participants' feelings about key factors relevant to positive emotional experiences.

4.1 What Participants Valued in the Existing Program

The program participants valued most being able to learn about their cultural ancestry. Several participants shared that one of the key motivators for participating in the program was to learn about the histories of some of their ancestors while also moving toward healthier habits. Therefore, participants shared that it would be important to keep the culturally relevant aspects of the program intact. P4 explained that she appreciated the program's emphasis on history, stating, "*I personally felt tied historically tied to like [–] I felt proud to be tied to like a history of like endurance and all of that. So the historical aspects were really a fighting point for me and I think, you know, as being a behavioral researcher, someone who like tries to understand like human thoughts and psychology and all that stuff. And, how it manifests in like our motivations and our health behaviors, you know, for me, like being tied to something that was bigger than myself was a real motivator, and it's also this racial-ethnic tie for me. I was very much tied to the historical story.*"

In addition to the cultural aspects, during the workshop, participants shared that they also valued the sense of community they felt as participants in the program. Participants shared that throughout the program, they would look for ways to help all members feel welcomed and included. P1 shared, "*For all the groups where*

I worked as a conductor [leader], we never made it a competition because people are on so many different levels. And we didn't want to make anybody feel uncomfortable, and even on the days that we walked, we'd normally have three groups of walkers; you had the really slow people and the medium people and the fast pacers, and you made everybody feel good about what they were doing. Because some people had real health challenges, and they wanted to improve, and you didn't want to discourage them."

Participants also valued the communal aspects of the program. They shared that it was important to focus not only on individual progress but progress as a team to motivate members. P2 explained, *"Hopefully, you would do some other exercise on your own, and you would report that exercise weekly to the conductor, and then there was that map that you showed earlier, that would show where, where the group was how many, you know how many miles the group walked, as well as the individual. So you wanted to help your team; you wanted to help your team get to 'freedom.'" Therefore, participants appreciated the cultural relevance but also the atmosphere that members of the program had built over time which fostered a sense of belonging and inclusiveness.*

4.2 Feedback on Mood Boards

When discussing the mood board, participants' values aligned majorly with their values in the existing program. Therefore, their opinions of the mood board designs seemed to align with their belief that the existing cultural and community components were motivators for participants. However, some design tensions emerged.

Participants commented that they valued the program's cultural aspects as it allowed them to learn about their ancestral history. Several participants shared that they liked the aesthetic of the *Historical* mood board for those reasons. In particular, participants liked that the board seemed to inspire feelings that aligned with their interest in learning about their past. The museum-like components emphasized a focus on history and learning. P5 shared, *"I'll just go ahead and halfway echo when [P4] was saying the historical feel sounds really interesting to me. I think maybe it's just my bias, but having a little bit of an educational perspective on health helps me get a little bit more intrigued by it. But, in trying to think broader from my peer group [younger adults] ... [they might prefer] anything that would have a more community connection."*

However, participants also shared that they felt the *Historical* mood might minimize the community aspects of the program because although museums can be social places, they tend to be experienced individually. Participants commented that while they might appreciate the historical context, they were unsure if it would promote feelings of communal support to the broader group and motivation to engage in physical activity. P4 discussed, *"I've seen that lovely portrait of Harriet Tubman a million times [in the mood board], and I love that portrait, but it doesn't make me want to walk all the time. [Laughs] If we can make it interactive, Harriet and then, you know, like, maybe that would be cool. And you know, 'Hey you go girl' ... But those pictures and just seeing that that old scrolly text, you know, which is beautiful, that makes me want to take a nap, not take a walk."*

Most participants, therefore, liked the *Gamify-Playful* board and felt that many components of the board aligned with the values

they had for the programs' community culture, and the use of gaming elements paired with cheerful colors could promote friendly encouragement and a cheerful mood for exercise. The playful colors, game-like characters, and focus on the journey aligned with what participants felt was promoting a sense of community members being on a health journey together. In addition, compared to the other two boards, participants felt the playful colors could be encouraging for getting people in the mood to exercise. P4 shared, *"I love history, but I feel like the historical ... When I think about exercising, it's not what gets me motivated, you know. So I kind of like the animated but not childlike cartoon, you know, like, something a little more relatable like I think we had it earlier on, like, show like kind of the Pixar ... not childlike cartoony ... I think that something that is little more relatable because through this process when we were talking I was just like, you know, it seems to me ... the people and interacting with people and interacting with each other, is kind of the secret sauce of [the program]. And so I would hate for the app to lose that."*

Participants did not like the *Darkness* mood. They agreed that there are many inspiring and empowering stories of those involved in the Underground Railroad, such as freedom seekers and abolitionists. However, they felt that reliving the conditions, while a very real part of history, could be triggering for some, leading to negative emotions.

4.3 Emotional Intentions: Factors Important for Positive Emotional Experiences

Based on the feedback we received, we found that emotions tied to the program centered around two overlapping aspects: **culture** and **community**. Participants appreciated learning about their ancestral history and valued learning about stories from the past. However, they also valued that the future app supports the community culture that the program had built over time. When discussing culture and community, several participants described several feelings they believed important to keep for a digital application to evoke positive emotional experiences among members.

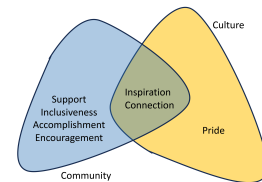


Figure 2: Emotions expressed during the workshop related to cultural and community aspects of the program. Two emotions, inspiration and connection, were discussed as relevant to both community and culture.

When discussing culture, participants used words such as **connection**, **inspiration**, and **pride** (Fig. 2). The use of these words aligned with their preference for opportunities to engage with historical facts (e.g., educational or museum-like experiences) conveyed in the *Historical* board. This was in contrast to reliving the experiences they felt were conveyed in the *Darkness* board.

Participants used words such as *support*, *inclusiveness*, *connection*, *inspiration*, *accomplishment*, and *encouragement* when discussing community. The use of these words aligned with their preference for experiences that support community building, positive motivation, and social engagement. Their preference for the *Gamify-Playful* board was aligned with these values as it included playful, cheerful colors and game-like motivators. Therefore, participants' preferences for the mood board designs included a mix of different elements of history and game-like, playful components they felt could better support positive, communal interactions in a digital space.

5 DISCUSSION

Similar to [37], our findings suggest that positive emotions beyond a pleasurable experience were important to the design of a culturally sensitive digital application to support African American heart health. We found that the positive emotions participants desired in the design fit under the categories of culture and community, two of the aspects of the original program participants valued most. Participants valued and wanted to keep in the digital application feelings of being inspired by past stories of the Underground Railroad, the connection it gave them to their ancestral history, and the pride they felt in reading about stories of resilience. In addition, participants valued what they described as the unique community experience of the existing program that provided members with support and encouragement, opportunities to connect on a shared journey, and inspiration (compared to the competition) to help each person accomplish their goals. Based on these findings, we discuss implications for designing culturally sensitive health technologies for African Americans that support positive emotion.

5.1 Balancing Mood in Culturally Sensitive Design

Mood, similar to emotion, can have different meanings but can play a significant role in influencing user experience [24]. Therefore, many researchers have studied the impact of mood on user experience [2, 27, 33]. For example, Wadley has explored ways technology can regulate mood [33], and Ashoori and colleagues have explored ways ambient lighting can affect mood in meeting spaces [2]. Mood boards are viewed as useful tools for engaging potential users or clients in discussions about user experience [8, 16] and have long been used in design practice [8]. In our study, we also found mood boards helpful as a tool to identify values and the emotions that participants wanted a digital application to evoke. However, we also uncovered a *need to carefully balance mood in the design of a culturally sensitive application*. We found that while certain aspects of the design were favored (history, learning) due to desires for cultural engagement, others were viewed to better support community engagement (gameplay, collaboration). Yet another *"experiencing the journey"* was viewed as potentially triggering due to the past traumatic historical events of that time. Therefore, aligned with prior work [2, 33], we found that exploring how design might influence mood was important to our design process, especially given the potentially varied emotions that subject matter might evoke. Through introducing the mood board as a design tool (e.g., probe [15]) we were able to identify positive emotions important to evoke

in the design of a future application: *support*, *inclusiveness*, *accomplishment*, *encouragement*, *inspiration*, *connection*, and *pride*. Some of these emotions and values have been found in prior work. For example, a person's feelings of collective responsibility for other community members' health (e.g., collectivism) have been found key for designing health applications supporting African Americans [12] and overlaps with findings from our study. We build on this and other prior work by identifying emotions key to the design of similar applications. Yet, we also highlight the need to balance and adjust to account for potentially negative emotions that might arise.

5.2 Exploring Intersectional Cultural Values

We found that design tensions that emerged related to different components of culture (e.g., shared ancestral culture and local community culture) that our participants value. Because participants believed that keeping both cultural experiences intact was important to successfully transitioning to a digital application, we were required to rethink our initial design ideas to attempt to support these different values. This finding was somewhat aligned with findings by Vigil-Hayes [32] that suggests that even among similar groups, cultural differences may arise due to nuances. Similarly, Grimes and colleagues [11] highlight the importance of designing "deeply local" applications. For us, this tailoring required us to focus on the layered needs and values of overlapping cultural experiences. Therefore, when designing applications to support communities with intersectional cultures, it can be useful for *designers to not only understand emotional intentions* [37] but also *gain a deeper understanding of the different cultural dimensions that contribute to overall user experience*. In our case, participants appreciated the emphasis on their ancestral histories and the unique values they had developed in the program.

5.3 Summary

We explored the relationship between emotion and design through the case of a culturally sensitive application to support African American heart health. We identified participants' values for both culture and community in an existing program and their desire for similar experiences when transitioning to a digital space. Using Yoon and colleagues' process [37] for identifying emotional intentions, we contribute values and emotions important to the design process. While our findings related to emotional design likely cannot be generalized [9], we discuss suggestions for approaching emotional design that may be transferable to other culturally sensitive health applications.

ACKNOWLEDGMENTS

We want to thank our community partners and those who contributed to the participatory design process. This work is funded by the National Institutes of Health Grant No. 1R21NR019702.

REFERENCES

- [1] [n. d.]. What is the Underground Railroad? - Underground Railroad (U.S. National Park Service). <https://www.nps.gov/subjects/undergroundrailroad/what-is-the-underground-railroad.htm>. Accessed: 2024-1-23.
- [2] Maryam Ashoori, Rachel K E Bellamy, and Justin D Weisz. 2015. Creating the Mood: Design for a Cognitive Meeting Room. In *Proceedings of the 33rd Annual*

- ACM Conference Extended Abstracts on Human Factors in Computing Systems (<conf-loc>, <city>Seoul</city>, <country>Republic of Korea</country>, <conf-loc>) (CHI EA '15). Association for Computing Machinery, New York, NY, USA, 2001–2006.
- [3] Natalie C Benda, Enid Montague, and Rupa S Valdez. 2020. Design for inclusivity. In *Design for Health*. Elsevier, 305–322.
 - [4] Scott Brave and Cliff Nass. 2007. Emotion in human-computer interaction. In *The human-computer interaction handbook*. CRC Press, 103–118.
 - [5] Laprincess C Brewer, Sarah Jenkins, Kandace Lackore, Jacqueline Johnson, Clarence Jones, Lisa A Cooper, Carmen Radecki Breitkopf, Sharonne N Hayes, and Christi Patten. 2018. mHealth Intervention Promoting Cardiovascular Health Among African-Americans: Recruitment and Baseline Characteristics of a Pilot Study. *JMIR Res. Protoc.* 7, 1 (Jan. 2018), e31.
 - [6] Laprincess C Brewer, Clarence Jones, Joshua P Slusser, Maarya Pasha, Mathias Lalika, Megan Chacon, Patricia Takawira, Stanton Shanedling, Paul Erickson, Cynthia Woods, Ashton Krogman, Daphne Ferdinand, Paul Underwood, Lisa A Cooper, Christi A Patten, and Sharonne N Hayes. 2023. mHealth Intervention for Promoting Hypertension Self-management Among African American Patients Receiving Care at a Community Health Center: Formative Evaluation of the FAITH! Hypertension App. *JMIR Form Res* 7 (June 2023), e45061.
 - [7] Rafael A Calvo, Dorian Peters, and Sidney D'Mello. 2015. When technologies manipulate our emotions. *Commun. ACM* 58, 11 (Oct. 2015), 41–42.
 - [8] Huang-Ming Chang, Marta Diaz, Andreu Català, Wei Chen, and Matthias Rauterberg. 2014. Mood Boards as a Universal Tool for Investigating Emotional Experience. In *Design, User Experience, and Usability: User Experience Design Practice*. Springer International Publishing, 220–231.
 - [9] John W Creswell and Cheryl N Poth. 2016. *Qualitative Inquiry and Research Design: Choosing Among Five Approaches*. SAGE Publications.
 - [10] Barney G Glaser. 1965. The Constant Comparative Method of Qualitative Analysis. *Soc. Probl.* 12, 4 (1965), 436–445.
 - [11] Andrea Grimes, Martin Bednar, Jay David Bolter, and Rebecca E Grinter. 2008. EatWell. In *Proceedings of the ACM 2008 conference on Computer supported cooperative work - CSCW '08* (San Diego, CA, USA). ACM Press, New York, New York, USA.
 - [12] Andrea Grimes and Rebecca E Grinter. 2007. Designing Persuasion: Health Technology for Low-Income African American Communities. In *Persuasive Technology*. Springer Berlin Heidelberg, 24–35.
 - [13] Jonna Häkikilä, Mikael Wiberg, Nils Johan Eira, Tapio Seppänen, Ilkka Juuso, Maija Mäkilä, and Katrin Wolf. 2020. Design Sensibilities - Designing for Cultural Sensitivity. In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society* (Tallinn, Estonia) (NordiCHI '20, Article 125). Association for Computing Machinery, New York, NY, USA, 1–3.
 - [14] Christina N Harrington and Anne Marie Piper. 2018. Informing Design through Sociocultural Values: Co-Creation with Low-Income African-American Older Adults. In *Proceedings of the 12th EAI International Conference on Pervasive Computing Technologies for Healthcare* (New York, NY, USA) (PervasiveHealth '18). Association for Computing Machinery, New York, NY, USA, 294–298.
 - [15] Hilary Hutchinson, Heiko Hansen, Nicolas Roussel, Björn Eiderbäck, Wendy Mackay, Bo Westerlund, Benjamin B Bederson, Allison Druin, Catherine Plaisant, Michel Beaudouin-Lafon, Stéphane Conversy, and Helen Evans. 2003. Technology probes. In *Proceedings of the conference on Human factors in computing systems - CHI '03* (Ft. Lauderdale, Florida, USA). ACM Press, New York, New York, USA.
 - [16] Andrés Lucero, Dzmitry Aliakseyeu, and Jean-Bernard Martens. 2008. Funky wall: presenting mood boards using gesture, speech and visuals. In *Proceedings of the working conference on Advanced visual interfaces* (Napoli, Italy) (AVI '08). Association for Computing Machinery, New York, NY, USA, 425–428.
 - [17] Aqueasha Martin-Hammond and Tanjala S Purnell. 2022. Bridging Community, History, and Culture in Personal Informatics Tools: Insights from an Existing Community-Based Heart Health Intervention for Black Americans. *Proc. ACM Hum.-Comput. Interact.* 6, GROUP (Jan. 2022), 1–23.
 - [18] John McCarthy and Peter Wright. 2004. Technology as experience. *Interactions* 11, 5 (Sept. 2004), 42–43.
 - [19] Sanjana Mendu, Mehdi Boukhechba, Janna R Gordon, Debajyoti Datta, Edwin Molina, Gloria Arroyo, Sara K Proctor, Kristen J Wells, and Laura E Barnes. 2018. Design of a Culturally-Informed Virtual Human for Educating Hispanic Women about Cervical Cancer. In *Proceedings of the 12th EAI International Conference on Pervasive Computing Technologies for Healthcare* (New York, NY, USA) (PervasiveHealth '18). Association for Computing Machinery, New York, NY, USA, 360–366.
 - [20] Enid Montague and Jennifer Perchonok. 2012. Health and wellness technology use by historically underserved health consumers: systematic review. *J. Med. Internet Res.* 14, 3 (May 2012), e78.
 - [21] Enid Montague, Woodrow Winchester, Rupa Valdez, Monifa Vaughn-Cooke, and Jennifer Perchonok. 2013. Considering Culture in the Design and Evaluation of Health IT for Patients. *Proc. Hum. Fact. Ergon. Soc. Annu. Meet.* 57, 1 (Sept. 2013), 1088–1092.
 - [22] Teresa K O'Leary, Elizabeth Stowell, Everlyne Kimani, Dhaval Parmar, Stefan Olafsson, Jessica Hoffman, Andrea G Parker, Michael K Paasche-Orlow, and Timothy Bickmore. 2020. Community-Based Cultural Tailoring of Virtual Agents. In *Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents* (Virtual Event, Scotland, UK) (IVA '20, Article 43). Association for Computing Machinery, New York, NY, USA, 1–8.
 - [23] Andrea G Parker and Rebecca E Grinter. 2014. Collectivistic health promotion tools: Accounting for the relationship between culture, food and nutrition. *Int. J. Hum. Comput. Stud.* 72, 2 (Feb. 2014), 185–206.
 - [24] Zhuochao Peng, Pieter M A Desmet, and Haian Xue. 2023. Mood in Experience Design: A Scoping Review. *She Ji: The Journal of Design, Economics, and Innovation* 9, 3 (Sept. 2023), 330–378.
 - [25] Jennifer Perchonok and Enid Montague. 2012. The Need to Examine Culture in Health Technology. *Proc. Hum. Fact. Ergon. Soc. Annu. Meet.* 56, 1 (Sept. 2012), 1847–1851.
 - [26] Tanjala S Purnell, Ina Glenn-Smith, Sunjae Bae, Grecia Vargas, Anika L Hines, and Jeanne Charleston. 2020. Abstract 09: The Health Freedom Path to Wellness Program: A Novel Educational and Behavioral Intervention to Promote Cardiovascular Health Equity. *Circulation* 141, Suppl_1 (March 2020), A09–A09.
 - [27] Leah M Scolare, Eric P S Baumer, Lindsay Reynolds, and Geri Gay. 2016. Building Mood, Building Community: Usage Patterns of an Interactive Art Installation. In *Proceedings of the 2016 ACM International Conference on Supporting Group Work* (Sanibel Island, Florida, USA) (GROUP '16). Association for Computing Machinery, New York, NY, USA, 201–212.
 - [28] Fadia T Shaya, Anna Gu, and Elijah Saunders. 2006. Addressing cardiovascular disparities through community interventions. *Ethn. Dis.* 16, 1 (2006), 138–144.
 - [29] Yaoxi Shi, Peter Koval, Vassilis Kostakos, Jorge Goncalves, and Greg Wadley. 2023. "Instant Happiness": Smartphones as tools for everyday emotion regulation. *Int. J. Hum. Comput. Stud.* 170 (Feb. 2023), 102958.
 - [30] Wally Smith, Greg Wadley, Oliver Daly, Marianne Webb, Jo Hughson, John Hajek, Anna Parker, Robyn Woodward-Kron, and David Story. 2017. Designing an app for pregnancy care for a culturally and linguistically diverse community. In *Proceedings of the 29th Australian Conference on Computer-Human Interaction* (Brisbane, Queensland, Australia) (OZCHI '17). Association for Computing Machinery, New York, NY, USA, 337–346.
 - [31] Wally Smith, Greg Wadley, Sarah Webber, Benjamin Tag, Vassilis Kostakos, Peter Koval, and James J Gross. 2022. Digital Emotion Regulation in Everyday Life. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (<conf-loc>, <city>New Orleans</city>, <state>LA</state>, <country>USA</country>, <conf-loc>) (CHI '22, Article 444). Association for Computing Machinery, New York, NY, USA, 1–15.
 - [32] Morgan Vigil-Hayes, Ann Futterman Collier, Shelby Hagemann, Giovanni Castillo, Keller Mikkelsen, Joshua Dingman, Andrew Muñoz, Jade Luther, and Alexandra McLaughlin. 2021. Integrating Cultural Relevance into a Behavioral mHealth Intervention for Native American Youth. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1 (April 2021), 1–29.
 - [33] Greg Wadley. 2016. Mood-enhancing technology. In *Proceedings of the 28th Australian Conference on Computer-Human Interaction* (Launceston, Tasmania, Australia) (OZCHI '16). Association for Computing Machinery, New York, NY, USA, 326–332.
 - [34] Greg Wadley, Vassilis Kostakos, Peter Koval, Wally Smith, Sarah Webber, Anna Cox, James J Gross, Kristina Höök, Regan Mandryk, and Petr Slovak. 2022. The Future of Emotion in Human-Computer Interaction. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22, Article 93). Association for Computing Machinery, New York, NY, USA, 1–6.
 - [35] Greg Wadley, Amanda Krause, Jiahui Liang, Zihe Wang, and Tuck Wah Leong. 2020. Use of music streaming platforms for emotion regulation by international students. In *Proceedings of the 31st Australian Conference on Human-Computer Interaction* (Fremantle, WA, Australia) (OZCHI '19). Association for Computing Machinery, New York, NY, USA, 337–341.
 - [36] Maria K Wolters, Zawadhafsa Mkulo, and Petra M Boynton. 2017. The Emotional Work of Doing eHealth Research. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (<conf-loc>, <city>Denver</city>, <state>Colorado</state>, <country>USA</country>, <conf-loc>) (CHI EA '17). Association for Computing Machinery, New York, NY, USA, 816–826.
 - [37] Jungkyoon Yoon, Anna E Pohlmeier, and Pieter M A Desmet. 2014. The mood street: designing for nuanced positive emotions. In *Proceedings of the 8th Nordic Conference on Human-Computer Interaction: Fun, Fast, Foundational* (Helsinki, Finland) (NordiCHI '14). Association for Computing Machinery, New York, NY, USA, 707–716.